KALOYAN DIMITROV

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EDUCATION

Yale University, New Haven, CT — Expected Graduation May 2029
B.S. Electrical Engineering (ABET-accredited)

EXPERIENCE

Computational Methods for Curious Robots Lab, *Undergraduate Researcher*, New Haven, Connecticut — Oct 2025 - PRESENT
- Developing simulations and novel control strategies, edge AI

Yale University, *Sr. Lab Technician*, New Haven, Connecticut — Oct 2025 - PRESENT
- Developing lab exercises and tools for Yale engineering courses

Yale Helix, *Fellow*, New Haven, Connecticut — Oct 2025 - Apr 2026
- Collaborating with startup PiroGon on medical device development for neurostimulation and EMG

Akhnaton LTD, *Automation Specialist*, Sofia, Bulgaria — Jul 2024 - Jun 2025
- Integrated a new collaborative robot into the product lineup

LEADERSHIP EXPERIENCE

Bulldogs Racing Team(Formula Student), New Haven, Connecticut — Sept 2025 – Apr 2026
- Founded and led software subteam to build full telemetry system and integrate with hardware

ACS Phoenix FIRST Robotics Competition Team, *Founder & Team Captain*, Sofia, Bulgaria — Nov 2022 – Mar 2025
- Founded and led the first Bulgarian FIRST Robotics team
- Led engineering, programming, fundraising, recruiting, and marketing, managing a $10,000 yearly budget

EnduroSat Space Challenges Bootcamp, *Team Leader*, Gela, Bulgaria — Aug – Sep 2022
- Led a 5-member team for 5 weeks to build a ground station, communicating with real LEO satellite
- Developed satellite tracking software and electronics for antenna guidance from scratch

PROJECTS

BLAST CubeSat, New Haven, Connecticut — Sept 2026 - PRESENT
- Built full SITL/HITL environment for testing
- Designed control strategy and implemented firmware for communications and control

PCHELICHKA, Sofia, Bulgaria — March 2024 - PRESENT
- Developed fully autonomous drone for pollination and secured start-up incubator and EU grant funding
- Came up with novel bio-inspired contactless pollination technique
- Implemented full controls and sensor system, plant disease detection AI algorithm, and dashboard

Paintball, Sofia, Bulgaria — Sept 2024 – Aug 2025
- Contracted by a local paintball arena chain to develop interconnected checkpoints for teambuilding games
- Designed electronics, mesh radio communication, firmware, and a web panel for game management

Swarm Guidance and Communication System, Sofia, Bulgaria — Nov 2021 – Oct 2024
- First author of Q2 journal publication
- Proposed and evaluated an algorithm for collaboration within robotic swarms with no centralized communication
- Conducted research in biology and engineering; built simulation; collected, analyzed, and visualized performance data

SHARO, Sofia, Bulgaria — Oct 2022 – Apr 2023
- Developed full software stack for 3D-printed quadruped from scratch (movement, teleoperation UI, and AI vision)
- Introduced novel method to describe limb trajectories with Bézier curves

Silica Gel and Xerogel Preparation, Sofia, Bulgaria — Sep 2023 –Jun 2024
- Studied the effects of Cl^- ions as a catalyst in silica gel and xerogel formation
- Designed and conducted experiment in wet lab; performed measurements, analyzed data and authored paper

SKILLS

- *Computer Skills:* HPC, AI(PyTorch, Jax); Web development; Embedded(C/C++, Rust); CAD; Electronics
- *Language Skills:* English (full working proficiency); German, Italian (elementary proficiency); Bulgarian (native)